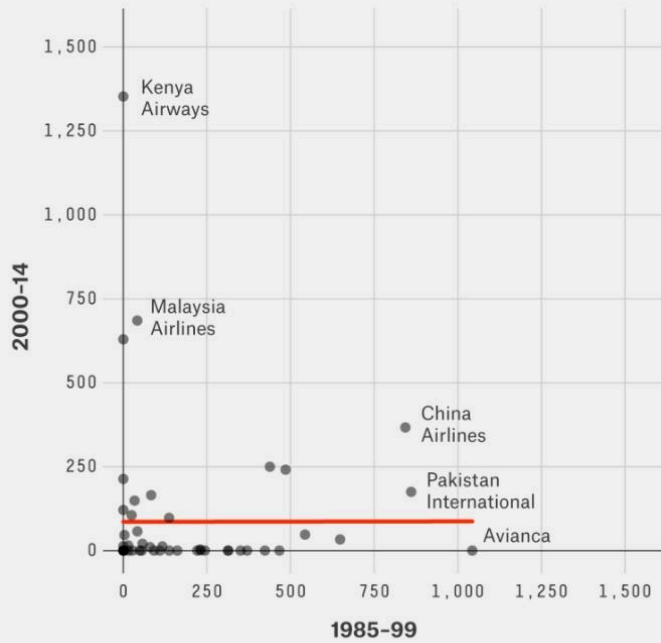


Fatalities by Airline Are Highly Unpredictable

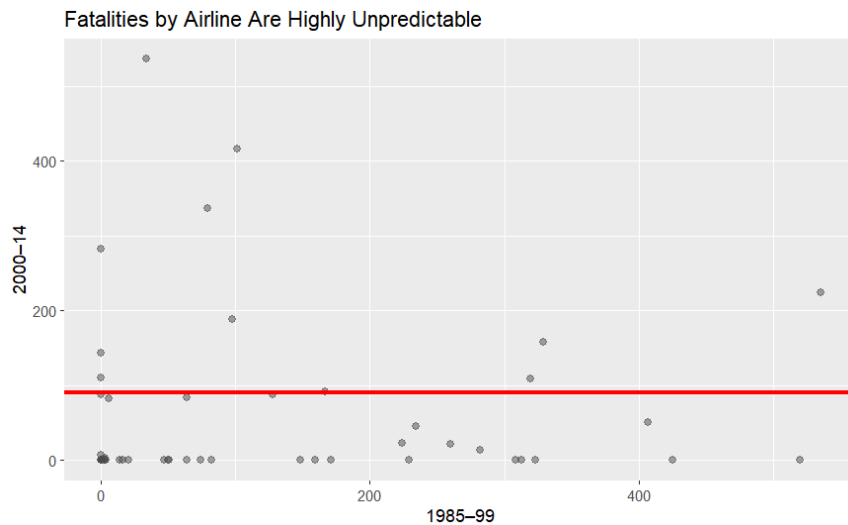
Fatalities adjusted for seats available and distance traveled
(deaths per 1 trillion seat kilometers)



Original Screenshot

My attempt:

```
ggplot(airline_safety, aes(x = fatalities_85_99, y = fatalities_00_14)) +  
  geom_point(color = "gray30", size = 2, alpha = 0.5) +  
  # Plot all airlines as points.  
  # I chose a dull gray color and partial transparency (alpha = 0.5)  
  # to mimic the understated look of the original graphic and avoid  
  # over-emphasizing any airline.  
  # Add horizontal reference line similar to the original  
  geom_hline(yintercept = 90, color = "red", linewidth = 1.2) +  
  # Add titles and axis labels.  
  labs(  
    title = "Fatalities by Airline Are Highly Unpredictable",  
    x = "1985–99",  
    y = "2000–14"  
  )
```



Specific challenges I couldn't overcome independently:

- Given dataset missing the original information
- Exact placement of the red reference line
- Airline labels

AI Attempt:

```
ggplot(airline_safety, aes(x = fatalities_85_99, y = fatalities_00_14)) +
```

```
# Plot airline points
```

```
geom_point(color = "gray30", size = 3, alpha = 0.8) +
```

```
geom_hline(yintercept = 90, color = "red", linewidth = 1.2) +
```

```
labs(
```

```
  title = "Fatalities by Airline Are Highly Unpredictable",
```

```
  subtitle = "Fatalities adjusted for seats available and distance traveled\n(deaths per 1  
trillion seat kilometers)",
```

```
  x = "1985–99",
```

```
  y = "2000–14"
```

```
) +
```

```
theme_minimal() +
```

```
theme(
```

```
  plot.title = element_text(size = 20, face = "bold"),
```

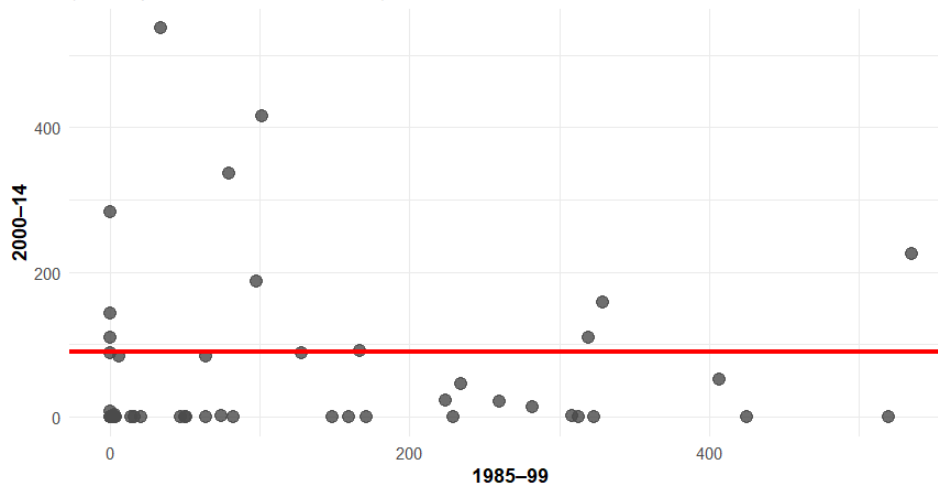
```
  plot.subtitle = element_text(size = 11),
```

```
  axis.title = element_text(face = "bold")
```

```
)
```

Fatalities by Airline Are Highly Unpredictable

Fatalities adjusted for seats available and distance traveled
(deaths per 1 trillion seat kilometers)



Phase 3:

After creating my initial attempt at recreating the visualization, I used AI tools to help solve several specific problems that I could not reproduce on my own. The original chart shows airline names next to certain points, a more refined theme, and a specific placement of the red reference line. My dataset and initial code were missing these elements.

Third, I asked AI about the exact placement of the red reference line. Because the dataset does not clearly state what value the line represents, AI explained that the original chart likely uses a visual reference point rather than a calculated statistic. As a result, the $yintercept = 90$ value I used is only an approximation.

Finally, AI confirmed an important limitation: many published charts like this one are finished using design software such as Adobe Illustrator after being created in statistical software. This means it may not be possible to perfectly replicate the exact label positions and visual refinements using only ggplot2.

AI Interactions:

1. Base Structure (Same in Both Charts)

Both charts begin with the same foundation:

```
«» R  
ggplot(airline_safety, aes(x = fatalities_85_99, y = fatalities_00_14))
```

This tells **ggplot2**:

- Use the dataset `airline_safety`
- Plot **fatalities from 1985–1999 on the x-axis**
- Plot **fatalities from 2000–2014 on the y-axis**

This creates a **scatter plot comparing airline safety across two time periods**. Each point represents one airline.

2. Improving the Points

Original version:

```
</> R  
geom_point(color = "gray30", size = 2, alpha = 0.5)
```

Improved version:

```
</> R  
geom_point(color = "gray30", size = 3, alpha = 0.8)
```

Changes made:

Change	Why it was done
size = 3 instead of 2	Makes the points easier to see
alpha = 0.8 instead of 0.5	Reduces transparency so the points stand out more

The color `gray30` was kept so the plot still looks neutral and professional.



Both charts include:

```
<> R
geom_hline(yintercept = 90, color = "red", linewidth = 1.2)
```

This adds a **horizontal reference line at 90 fatalities**.

Purpose:

- Helps highlight airlines that had **very high fatality rates in 2000–2014**
- Creates a visual threshold so viewers can quickly see which airlines exceed it.

4. Adding a Subtitle

Original chart:

```
<> R
labs(
  title = "Fatalities by Airline Are Highly Unpredictable",
  x = "1985-99",
  y = "2000-14"
)
```

Improved chart:

```
<> R
labs(
  title = "Fatalities by Airline Are Highly Unpredictable",
  subtitle = "Fatalities adjusted for seats available and distance traveled\n(deaths per 1 trilli",
  x = "1985-99",
  y = "2000-14"
)
```

Why this was added:

- The subtitle explains what the fatality metric actually represents
- The `\n` creates a line break, making the subtitle easier to read.

5. Improving the Theme

The improved version adds:

```
</> R
theme_minimal()
```

This creates a clean background with minimal grid lines, which:

- Reduces visual clutter
- Makes the data points easier to focus on.

6. Enhancing Text Hierarchy

The improved chart also adds:

```
</> R
theme(
  plot.title = element_text(size = 20, face = "bold"),
  plot.subtitle = element_text(size = 11),
  axis.title = element_text(face = "bold")
)
```

These changes improve visual hierarchy:

Element	Change	Purpose
plot.title	Bigger and bold	Makes the main message stand out
plot.subtitle	Slightly smaller	Provides supporting context
axis.title	Bold	Makes axis labels clearer

This helps viewers understand the chart faster, which is especially important for an executive audience.

Final Product:

```
ggplot(airline_safety, aes(x = fatalities_85_99, y = fatalities_00_14)) +  
  
  # Plot airline points  
  geom_point(color = "gray30", size = 3, alpha = 0.8) +  
  
  # Horizontal reference line  
  geom_hline(yintercept = 90, color = "red", linewidth = 1.2) +  
  
  # Titles and labels  
  labs(  
    title = "Fatalities by Airline Are Highly Unpredictable",  
    subtitle = "Fatalities adjusted for seats available and distance traveled\n(deaths per 1  
trillion seat kilometers)",  
    x = "1985–99",  
    y = "2000–14"  
  ) +  
  
  # Styling to match the clean design of the original  
  theme_minimal() +  
  theme(  
    plot.title = element_text(size = 20, face = "bold"),  
    plot.subtitle = element_text(size = 11),  
    axis.title = element_text(face = "bold")  
  )
```


Phase 4: Reflection

The most challenging part of recreating the visualization was matching the overall design and small details of the original chart. While it was fairly straightforward to plot the points using `ggplot()`, it was harder to replicate the styling elements that make the original chart look polished. For example, I struggled with matching the exact appearance of the reference line, the point sizes, and the text formatting. Another challenge was the airline labels shown on the original graphic. My dataset did not include clear instructions for labeling specific airlines, and placing text labels on a scatterplot without overlap can be difficult.

To approach these challenges, I first focused on recreating the core structure of the visualization: plotting the two variables and adding the reference line. After that, I gradually improved the design by adjusting colors, transparency, and themes. I experimented with different sizes and alpha values to make the points look closer to the original chart.

Through this process, I learned several useful visualization techniques in `ggplot2`. I became more comfortable using functions like `geom_point()` to control point appearance and `geom_hline()` to add reference lines that highlight important values. I also learned how to improve the visual hierarchy of a plot using `theme_minimal()` and the `theme()` function to adjust title size, subtitles, and axis labels.

AI assistance was helpful for refining the visualization after I created the basic plot. It helped me understand how to improve the styling and suggested ways to structure the plot code more clearly. However, AI could not perfectly reproduce the original design, especially when it came to exact label placement and design details that are often added in graphic design software after the chart is created.